

Exercise 5

NULL FUNCTIONS

Table 1 Employees

emp-id	Name	Department	Salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

Q1: Show all employees with their salary. If salary is NULL then display 0.

```
SELECT employee_id, Name,  
       IF NULL(salary, 0) AS salary-with-default  
FROM Employees
```

Employee-id	Name	Salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2 : Show employee Name with their departments. If department is NULL Show "NOT assigned"

```
SELECT employee_id, Name,
       IF NULL (department, "NOT assigned")
       AS department_name
FROM Employees
```

emp-id	Name	Department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2 : ORDERS

Order-id	Customer-id	Delivery-date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

Q3 : Find orders with NULL

```
SELECT order_id, customer_id
FROM ORDERS
```

```
WHERE customer_id IS NULL
```

Order-id	Customer-id
103	NULL

Q4: Show all orders, if delivery date is NULL show Pending

```
SELECT order-id, customer-id,  
       IF NULL (delivery-date, 'Pending') AS  
       delivery status  
FROM ORDERS
```

Order-id	Customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: STUDENTS

Student-id	Name	Grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

Q5: Show all students and grades, replace NULL with 0

```
SELECT student-id, name, final-grade  
       IF NULL (Grade, 0) AS Final-grade  
FROM STUDENTS
```

Student-id	Name	Final_Grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6 : Count students who haven't been graded

SELECT COUNT(~~Att~~^{*}) AS NOT-Graded-count
FROM Students
WHERE Grade IS NULL

Not-Graded-count
1

Table 4 : PRODUCTS

product-id	Name	Price	Discount
S01	Keyboard	25	NULL
S02	Mouse	15	5
S03	Monitor	100	NULL

Final price after

Q7 : Show all and if discount is NULL
then 0.

SELECT product-id, name, ~~price~~
IF NULL (discount, 0) AS final price
FROM PRODUCTS

product-id	Name	Price	final-price
S01	Keyboard	25	0
S02	Mouse	15	5
S03	Monitor	100	0

Table 5: CUSTOMERS

customer-id	Name	Email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Nia	NULL

Q8: Count how many customers have no mail

SELECT COUNT(~~NULL~~*) AS missing-email-count
FROM CUSTOMERS
WHERE Email IS NULL

missing-email-count
2

Q9: Show all customers with email, if NULL display "NO Email"

SELECT customer_id, name
IF NULL(email, 'NO email') AS email-display
FROM CUSTOMERS

customer-id	Name	Email-display
1	Linda	No email
2	Joseph	joseph@email.com
3	Nia	No email

Table 6 : Payment

payment-id	method	status
301	Credit	NULL
302	PayPal	Success
303	NULL	Failed

Q10 : show payment details with method replaced by 'Unkown' if NULL

```

SELECT payment-id, status
       IF NULL (method, 'Unkown') AS
       method-display
FROM   Payment
  
```

pay-id	method-display	status
301	Credit	NULL
302	PayPal	Success
303	Unkown	Failed

Table 7 : INVENTORY

Q11

item_id	item_name	quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

Q11 : show items and their quantity
(0 if NULL)

```
SELECT item_id, item_name, IF NULL(quantity, 0)
      AS quantity-checked
FROM INVENTORY;
```

item_id	item_name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8 : Employees - Extra

emp-id	bonus	commission
1	NULL	300
2	100	NULL
3	NULL	NULL

Q12 : Show emp-id and first available value

```
SELECT emp-id, COALESCE (bonus, commission, 0)
      AS first-available-reward
FROM   Employees - Extra ;
```

emp-id	first-available-reward
1	300
2	100
3	0

Table 9 : Classes

Class_id	subject	room
11	Math	NULL
12	Science	Lab A
13	English	NULL

Q13: Count classes with no assigned room

```
SELECT COUNT(NULL*) AS no-room-count
FROM Classes
WHERE room IS NULL;
```

no-room-count
2

Table 10 : Attendance

Student_id	Date	Status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Present

Q14: Show attendance record with status, Replace NULL with 'Not marked'

```
SELECT student_id, date
      IF NULL(status, 'Not Marke')
FROM Attendance;
```

Table 11 : Bank_Accounts

account-id	account-type	balance
A1	Savings	NULL
A2	Cheque	5000
A3	NULL	2000

Q15 : Show account ID, account type (or 'unkown') and balance (or 0)

```
SELECT account_id, IF NULL (account-type,
                             'Unkown'), IF NULL (balance, 0)
FROM   Bank - Accounts ;
```

account-id	account-type	balance
A1	Savings	0
A2	Cheque	5000
A3	Unkown	2000

Table 12 : Projects

project-id	title	start-date	end-date
1	Website revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

Q16 : Show all projects with a start date (IF NULL THEN 'TBD')

```
SELECT project-id, title,
       IF NULL (start-date, 'TBD') AS start-  
FROM Projects ;
```

project-id	title	start-display
1	Website revamp	TBD
2	Mobile App	2025-06-01
3	Data Migration	TBD

Table 13 : Reviews

review-id	product-id	comment	Rating
1	S01	Great product	4
2	S02	NULL	NULL
3	S03	Works fine	3

Q17 : Display reviews showing comment (or 'NO comment') and rating (or 0)

```
SELECT review-id, product-id,
       IF NULL (comment, 'NO comment') AS comment,
       display,
       IF NULL (rating, 0) AS rating-display
FROM   Reviews ;
```

review-id	product-id	Comment ^{display}	Rating-display
1	S01	Great product	4
2	S02	No comment	0
3	S03	Works fine	3

Table 14 : Suppliers

Supplier-id	Name	phone	alt-phone
1	Global Goods	NULL	123456789
2	Bestsupplies	987654321	NULL
3	ValueSource	NULL	NULL

Q18: Show supplier-id, name and contact number

```
SELECT Supplier-id,
       Name,
       COALESCE (phone, alt-phone, 0) AS
       Contact-number
FROM Suppliers;
```

Supplier-id	Name	Contact-number
1	Global Goods	123456789
2	Best supplies	987654321
3	ValueSource	0

Table 15 : USER-SETTINGS

user-id	theme	language	timezone
1	NULL	English	NULL
2	Dark	NULL	UTC + 1
3	NULL	NULL	NULL

Q19 : Show all users and their preference.

Replace NULLS

- Theme → "light"
- Language → "English"
- Timezone → "UTC"

```
SELECT user-id,
IF NULL (theme, light) AS theme-set
IF NULL (language, English) AS language-set
IF NULL (timezone, UTC) AS timezone-set
FROM 'User-settings';
```

User-id	Theme-set	language-set	timezone-set
1	light	English	UTC
2	Dark	English	UTC + 1
3	light	English	UTC

Table 16 : Maintenance

record-id	machine-id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

- Q20 : Show maintenance log with
- issue → 'Unknown Issue'.
 - technician → 'Not Assigned'

```
SELECT record-id, machine-id,
IF NULL (issue, 'Unknown issue') AS
issue-log,
IF NULL (technician, 'Not Assigned')
AS technician_name
FROM Maintenance ;
```

record-id	machine-id	issue-log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex